

Zhihan Lei

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[Github](#) | [Google Scholar](#) | [LinkedIn](#)

EDUCATION

Carnegie Mellon University

Master of Science in Machine Learning (GPA: 3.7)

Emory University

Bachelor of Science in Physics, Minor in Computer Science (GPA: 3.8)

Pittsburgh, PA

Dec 2026

Atlanta, GA

Dec 2023

PUBLICATIONS & MANUSCRIPTS

Published:

Hu, Y., Lei, Z., et al. GRAG: Graph Retrieval-Augmented Generation. NAACL 2025, second author; 313 citations;

Hu, Y., Lei, Z., et al. CG-RAG: Research Question Answering by Citation Graph Retrieval-Augmented LLMs. SIGIR 2025, second author; 22 citations.

Under Review:

Lei, Z., et al. Inducing Reasoning Primitives from Agent Traces. NeurIPS 2026, first author; [arXiv](#)

Lei, Z., et al. Learning to Construct Practical Agentic Systems. NeurIPS 2026, co-first author; [arXiv](#)

Wang, B., Lei, Z., et al. SpecDrop: Parameter-Free Category-Conditioned Routing for Modular Specialization. NeurIPS 2026, second author.

RESEARCH EXPERIENCE

Carnegie Mellon University — LLM Agents & Reasoning

Graduate Researcher (Advisor: Prof. William Cohen)

Pittsburgh, PA

Jan 2026 - Present

- Proposed Reasoning Primitive Induction (first-author, NeurIPS 2026 submission): a self-improvement method where an LLM agent mines its own successful traces into reusable tools it invokes at test time — a gradient-free, test-time route to agent gains, complementary to RL post-training.
- Induced libraries beat their own source ReAct agent (DeepSeek-V3) — RuleArena NBA 30%→74% (+44pp), +30pp MuSR, +22pp NatPlan (paired-Δ 95% bootstrap CIs strictly positive); beat prior agent-memory baseline (AWM) at ~24% lower inference cost.
- Ablation: replacing tool descriptions with adversarial text caused 20–32pp drops on content-critical tasks but no degradation on structure-only ones; cross-judge agreement 96.6%.
- Co-first-authored a companion cross-benchmark study (NeurIPS 2026 submission) mapping when induction helps across 19 reasoning tasks: gains scale with baseline-tool weakness, e.g. +8.2pp on MedCalc at ~60% fewer input tokens.

Emory Machine Learning Lab — Graph Learning & Retrieval

Researcher (Advisor: Prof. Liang Zhao) | [Github](#)

Atlanta, GA

Apr 2024 - Feb 2025

- GRAG (NAACL 2025): divide-and-conquer subgraph retrieval at linear time with soft pruning, dual text/graph-view prompting; SOTA on multi-hop graph reasoning.
- CG-RAG (SIGIR 2025): chunk-level citation graph + Lexical-Semantic Graph Retrieval entangling sparse/dense via message passing; proved hybrid retrieval is a special case.

PROFESSIONAL EXPERIENCE

Amazon

Software Engineer Intern

Seattle, WA

May 2026 - Aug 2026

- Built an agentic analytics compiler that removes manual Redshift→PartiQL migration: an LLM agent transpiles queries, validates semantic correctness by executing against benchmark cases, and auto-generates dashboards from context hints.

ByteDance

Machine Learning Engineer Intern

Beijing, China

Jul 2023 - Sep 2023

- Fine-tuned face-matching model for account security: 62% → 83% precision; deployed as real-time identity verification.
- Modeled millions of daily user behavioral sequences (Spark SQL) for scammer detection; surfaced 10k high-risk users.

PROJECTS

SpecDrop: Modular Specialization | CMU | NeurIPS 2026 (Under Review)

Feb 2026 - May 2026

- Parameter-free category-conditioned routing: repurposes dropout to drive modular specialization with no learned gating network; stability-guided freezing locks converged modules to cut training compute.
- Designed the specialization metrics (module-category MI, pruning sensitivity) and per-domain eval; validated on CIFAR-100 ResNets and 60–125M MoE LMs.

SKILLS

Research: LLM Agents & Reasoning, Test-Time Compute, Self-Improving Agents, Tool Synthesis & Skill Discovery, LLM Evaluation, Modular Specialization (MoE), Retrieval-Augmented Generation, Graph Neural Networks

Methods: Deep Learning, Transformer/Attention, NLP, Fine-tuning, Probabilistic Graphical Models, Bayesian Inference

Languages & Tools: Python (PyTorch, Hugging Face, NumPy, Pandas), C++, SQL; CUDA, AWS, Spark, Docker, Git, Linux